Integration Step

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Hypothesis :

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\delta\Pi^- - \mathrm{d}\Gamma^*_x\Pi^-_x)_{\beta r}(x)\right|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|}, \tag{1}$$

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\mathrm{d}\Gamma^*_x)^\gamma_\beta\right|^q)^{\frac{1}{2}} \lesssim R^{\frac{D}{2}+|\beta|-|\gamma|} \text{ for populated } \gamma \neq \mathrm{pp}, \tag{2}$$

$$(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} \left|(\mathrm{d}\Gamma^*_{x+y}\Pi^-_{x+y} - \mathrm{d}\Gamma^*_x\Pi^-_x)_{\beta r}(x+y)\right|^q)^{\frac{1}{2}} \lesssim r^{2\alpha}(r+|y|)^{\alpha+\frac{D}{2}}(r+|y|+R)^{|\beta|-2-3\alpha}, \tag{3}$$

$$L\Pi_{x\gamma} = \Pi^-_{x\gamma} \bmod \mathcal{P}_{\leq|\gamma|-2} \tag{4}$$

$$L\delta\Pi_\beta = \delta\Pi^-_\beta \bmod \mathcal{P}_{\leq|\beta|-2} \tag{5}$$

$$\partial^{\mathbf{n}}((\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_\beta)(x) = 0,\ \forall x \in \mathbb{R}^d, \forall\, |\mathbf{n}| < 2+s \tag{6}$$

Goal :

$$\left(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}} |(\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_{\beta r}(x)|^q\right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-|\alpha|} \tag{7}$$

Lemma 1 (Liouville)**.** *Let g be a smooth function, $m \in \mathbb{N}$ and $\kappa \in \mathbb{R}$ with $\kappa < m$, $x \in \mathbb{R}^n$ a base point. Then there exists at most on couple (u,p), with u a smooth function and p a polynomial, such that :*

$$\begin{cases} L(u) = g - p \\ \deg(p) \leq m-2 \\ \forall\, |\mathbf{n}| \leq m, \partial^{\mathbf{n}}u(x) = 0. \limsup_{|x-y|\to\infty} \frac{u(y)}{|x-y|^\kappa} < \infty. \end{cases}$$

LZ: Si P est le polynôme de Taylor à l'ordre m centré en x de $L^{-1}g$, alors $u = L^{-1}g - P$ et $p = LP$. Dans l'approche de Martin on travaille sur P, jamais sur p. Il me semble que pour Felix c'est l'opposé. Je ne comprends pas bien l'utilité de considérer p à la place de P. GDB: Effectivement, et l'hypothèse sur les noyaux régularisant préservant les polynômes permet de passer de l'un à l'autre. Dans l'article de Felix, ça permet d'écrire l'edp Lu = g mod polynome de degré…ce qui est pratique, je n'ai pas d'autres explication pour l'instant.

Proof. Let $|\mathbf{n}| \leq m-2$.

$$0 = \partial^{\mathbf{n}}(L(u))(x) = \partial^{\mathbf{n}}g(x) - \partial^{\mathbf{n}}p(x),$$

so that p is determined by g. Then, let u and $\tilde{u}$ be two candidate and denote $w = u - \tilde{u}$. We have $L(w) = 0$, and $\limsup_{|x-y|\to\infty} \frac{w(y)}{|x-y|^\kappa} < \infty$. By Liouville's theorem, w is a polynomial with degree at most κ. Then $\forall\, |\mathbf{n}| \le m, \partial^{\mathbf{n}} w(x) = 0$ shows that $w = 0$. □

In view of (4) and (5), by linear combination we have $L(\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta) = \delta\Pi_\beta^- - (\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta \bmod \mathcal{P}_{<|\beta|+\frac{D}{2}-2}$. But it is not sufficient to caracterize $\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta$ in terms of $\delta\Pi_\beta^- - (\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta$, because the growth at infinity of $\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta$ is too big. Indeed, $(\mathrm{d}\Gamma_x^*\Pi_x)_\beta = \sum_{|\gamma|<|\beta|+\frac{D}{2}} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$. Now $\Pi_{x\gamma}(y)$ is expected to grow like $|x-y|^\gamma$. So the sum includes terms that grow almost like $2 + \frac{D}{2}$. In contrast, the vanishing order of $\delta\Pi_\beta - (\mathrm{d}\Gamma_x^*\Pi_x)_\beta$ at x would be $\alpha + \frac{D}{2} (= 2 + s)$, which is lower by more than 2 (we are in the regime $\alpha < 0$). This is the reason why we truncate the sum at order $2 + s$. LZ: Tout ça n'est pas très clair. Je vois dans [BOT] la restriction $|\mathbf{n}| < 2 + s$ dans la formule (161) qui définit $\mathrm{d}\Gamma_x^*$, mais ensuite la triangularité de $\mathrm{d}\Gamma_x^*$ est $|\gamma| < |\beta| + \frac{D}{2}$ (175). Je vois que tu fais référence à la discussion de [BOT, section 2.15]. On peut expliquer mieux ? GDB: Ici, les $(\mathrm{d}\Gamma_x^*)_\gamma^\beta$ sont simplement des coefficients constants, et ne participe pas à la croissance en l'infini, la triangularité est importante parce qu'elle définit l'ensemble d'indice de sommation. On est en gros en train de dire $(\mathrm{d}\Gamma_x^*\Pi_x)(y) \approx_{y\to\infty} C\Pi_{x\gamma_0}(y) \approx C\,|x-y|^{|\gamma_0|}$, où γ_0 est l'indice d'homogénéité maximale dans la somme : $|\gamma_0|$ presque égal à $|\beta| + \frac{D}{2}$. Et sans restriction sur β, $|\beta|$ peux aller jusqu'à 2. Furthermore, because $L\Pi_{x\beta} = \Pi_{x\beta}^- \bmod \mathcal{P}_{<|\beta|-2}$ and $L\delta\Pi_\beta = \delta\Pi_\beta^- \bmod \mathcal{P}_{<|\beta|-2}$, we restrict ourselves to $|\beta| < \lceil 2 + s \rceil$. As a result, we have the following

$$L(\delta\Pi_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}) = \delta\Pi_\beta^- - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}^- \quad \bmod \mathcal{P}_{<s} \tag{8}$$

and the hypothesis of the lemma are verified, so $U_x := \delta\Pi_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$ is completely determined by $F_x := \delta\Pi_\beta^- - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}^-$. LZ: Essayons de vérifier cela précisément. J'imagine que $m = \lceil 2 + s \rceil$ et que $\partial^{\mathbf{n}} u(x) = 0$ pour tout $|\mathbf{n}| \le m$ suit de [BOT, (164)]. Le contrôle à l'infini suit de ???

Remark 2. *The threshold $2 + s$ is too strong for $(\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta$, but we will need to deal with the remainder $\sum_{|\gamma|\ge 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}^-$ and see that this threshold makes sense. LZ: Not very clear, I guess this is again the previous discussion or [BOT, section 2.15].GDB: En fait je regrette cette remarque qui n'est pas vraiment vraie, je sais pas pourquoi j'ai écrit ça.*

We claim that (2) implies

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\sum_{|\gamma|\ge 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}^-)_r(x) \right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}-2} (r+R)^{|\beta|-|\alpha|}, \tag{9}$$

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\sum_{|\gamma|\ge 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}} (r+R)^{|\beta|-|\alpha|}. \tag{10}$$

So that by triangle inequality, we can add to our set of hypothesis

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(F_x)_r(x)|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}-2} (r+R)^{|\beta|-|\alpha|}, \tag{11}$$

and it suffices to show

$$\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(U_x)_r(x)|^q \right)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}} (r+R)^{|\beta|-|\alpha|}.$$

The strategy now is to take advantage of the uniqueness of solution to $Lu = F_x \bmod \mathcal{P}_{<s}$, to find another expression of U_x, that verifies (7). The germ (F_x) is $(2\alpha, \alpha+\frac{D}{2}-2)$-coherent LZ: why 2α instead of $\alpha-2$? Anyway, where is this proved? I looked into [BOT] but I could not identify the relevant estimate GDB: Sorry I forgot to justify this. I wrote it with estimate (203) of [BOT] in mind, which I added in the hypothesis :(3), but it is the coherence for the non-truncated germ. I am trying to prove it for the truncated one, and I beleive it is true : it seems we just remove terms that we have to estimate in the proof of the article, but it is not so obvious to write.GDB: edit lundi 19 : J'ai écrit les détails de la cohérence ci-dessous. and $(\alpha+\frac{D}{2}-2)$-homogeneous LZ: That is true if the reconstruction of F is 0, is that true here?.GDB: I beleive it is clearer now as it is just equation (11) above. Let us justify the coherence. In the article, we have $(2\alpha, \alpha+\frac{D}{2}-2)$-coherence for the untruncated germ : it is equation (203) [2]. Let's compute the increment for our truncated germ :

$$\begin{aligned}
F_{x+y} - F_x &= \sum_{|\gamma|<2+s} \left((\mathrm{d}\Gamma^*_{x+y})^\gamma_\beta \Pi^-_{x+y\gamma} - (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma} \right) \\
&= \sum_{|\gamma|<2+s} \left((\mathrm{d}\Gamma^*_{x+y})^\gamma_\beta \Pi^-_{x+y\gamma} - (\mathrm{d}\Gamma^*_x)^\gamma_\beta \sum_\theta (\Gamma^*_{xx+y})^\theta_\gamma \Pi^-_{x+y\theta} \right) \\
&= \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_{x+y})^\gamma_\beta \Pi^-_{x+y\gamma} - \sum_\theta \left((\mathrm{d}\Gamma^*_x \Gamma^*_{xx+y})^\theta_\beta + \sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta (\Gamma^*_{xx+y})^\theta_\gamma \right) \Pi^-_{x+y\theta} \\
&= \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_{x+y} - \mathrm{d}\Gamma^*_x \Gamma_{xx+y})^\gamma_\beta \Pi^-_{x+y\gamma} + \sum_\theta \sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta (\Gamma^*_{xx+y})^\theta_\gamma \Pi^-_{x+y\theta}
\end{aligned}$$

The first sum is treated in the article : the first factor is estimated in (195) [2] and the second is (141) [2], by Hölder's inequality and triangle inequality, we obtain the correct bound $r^{2\alpha}(r+|y|)^{\alpha-2+\frac{D}{2}}(r+|y|+R)^{|\beta|-\alpha}$. For the second sum, we use (2), (142) and (141), again with triangle inequality and Hölder's inequality, we get the estimate :

$$\begin{aligned}
\sum_\theta \sum_{|\gamma|\geq 2+s} R^{\frac{D}{2}+|\beta|-|\gamma|} |y|^{|\gamma|-|\theta|} r^{\theta-2} &\lesssim r^{2\alpha} \sum_{|\gamma|\geq 2+s} R^{\frac{D}{2}+|\beta|-|\gamma|} \sum_\theta |y|^{|\gamma|-|\theta|} r^{\theta-2-2\alpha} \quad (|\theta| \geq 2(\alpha+1) \text{ by (206)}) \\
&\lesssim r^{2\alpha} \sum_{|\gamma|\geq 2+s} R^{\frac{D}{2}+|\beta|-|\gamma|} (r+|y|)^{|\gamma|-2-2\alpha} \\
&\lesssim r^{2\alpha} (r+|y|)^{\alpha+\frac{D}{2}-2-2\alpha} \sum_{|\gamma|\geq 2+s} R^{\frac{D}{2}+|\beta|-|\gamma|} (r+|y|)^{|\gamma|-(\alpha+\frac{D}{2})} \\
&\lesssim r^{2\alpha} (r+|y|)^{\frac{D}{2}-2-\alpha} (r+|y|+R)^{|\beta|-\alpha}
\end{aligned}$$

Let K be the heat kernel, which is assumed to be a 2-regularising kernel (we would need to truncate it, for it to be compactly supported, but it is easy to deal with the smooth remainder). Define $\tilde{U}_x := \mathcal{K}(F)_x = K * (F_x) - T^{2+s}_x (K * (F_x))$. By the Schauder estimate theorem for coherent germs (Theorem 3.17 of [1]), $\tilde{U}_x$ is a well-defined $((2\alpha+2)\wedge 0, \alpha+\frac{D}{2})$-coherent and $\alpha+\frac{D}{2}$-homogeneous germ. Plus its homogeneity norm is dominated by the sum of the homogeneity and coherence norm of (F_x). This means that

$$\begin{aligned}
\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| (\tilde{U}_x)_r(x) \right|^q \right)^{\frac{1}{2}} &\lesssim r^{\alpha+\frac{D}{2}} ((C+R)^{|\beta|-\alpha} + (C+R)^{|\beta|-3\alpha-2}) \qquad (12) \\
&\lesssim r^{\alpha+\frac{D}{2}} (C+R)^{|\beta|-\alpha} (\text{because } 3\alpha+2 > \alpha) \qquad (13)
\end{aligned}$$

which is exactly the estimate we want to prove for (U_x).

Let's show that $\tilde{U}_x = U_x$. It suffices to check that both $\tilde{U}_x$ and U_x verifies Liouville's lemma's hypothesis. First, $L\tilde{U}_x = L(K * (F_x)) - L(T_x^{2+s}(K * (F_x))) = F_x +$ polynomial of degree $< s$. Then,

- The jet condition is verified by construction of $\tilde{U}_x$.
- With (10), we claimed that the remainder $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$ has vanishing order $\alpha + \frac{D}{2} = 2 + s$ at x, and together with (6), it shows that the jet condition is also verified for U_x.
- We already dealt with the growth condition of U_x, it was the reason we truncated.
- Growth condition for $\tilde{U}$? LZ: we do not study this in Martin's approach

which concludes.

It remains to prove that (2) implies (10) and (9). This is done purely by induction.

The sums $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi_{x\gamma}$ and $\sum_{|\gamma|\geq 2+s}(\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma}$ is restricted to populated γ that are not purely polynomial : else $\Pi^-_{x\gamma} = 0$, and $(\mathrm{d}\Gamma_x^*)_\beta^\gamma = (\mathrm{d}\Gamma_x^*)_\beta^{\delta_{\mathbf{n}}} = 0$ because $|\mathbf{n}| \geq 2 + s$. Furthermore, $|\gamma|_\prec < |\beta|_\prec$ so the induction hypothesis on $\Pi^-_{x\gamma}$ and $\Pi_{x\gamma}$ are available.

$$
\begin{aligned}
&\left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| \Big(\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma} \Big)_r(x) \right|^q \right)^{\frac{1}{2}} \\
&\leq \sum_\gamma \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} \left| ((\mathrm{d}\Gamma_x^*)_\beta^\gamma \Pi^-_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \text{(triangle inequality)} \\
&\leq \sum_\gamma \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{2q}} \left| (\mathrm{d}\Gamma_x^*)_\beta^\gamma \right|^{2q} \right)^{\frac{1}{2}} \left(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{2q}} \left| (\Pi^-_{x\gamma})_r(x) \right|^q \right)^{\frac{1}{2}} \text{(Hölder's inequality)} \\
&\leq \sum_{2+s\leq|\gamma|\leq|\beta|+\frac{D}{2}} R^{\frac{D}{2}+|\beta|-|\gamma|} r^{\gamma-2} \\
&\lesssim r^{\alpha-2+\frac{D}{2}} (r+R)^{|\beta|-\alpha}.
\end{aligned}
$$

And we proceed the same way for (10).

References

[1] Lucas Broux, Francesco Caravenna, and Lorenzo Zambotti. "Hairer's Multilevel Schauder Estimates without Regularity Structures". In: *Transactions of the American Mathematical Society* (Aug. 9, 2024). ISSN: 0002-9947, 1088-6850. DOI: 10.1090/tran/9245. arXiv: 2301.07517 [math.AP]. URL: http://arxiv.org/abs/2301.07517 (visited on 06/16/2026).

[2] Lucas Broux, Felix Otto, and Markus Tempelmayr. "Lecture Notes on Malliavin Calculus in Regularity Structures". In: *Stochastics and Partial Differential Equations: Analysis and Computations* (Sept. 9, 2025). ISSN: 2194-0401, 2194-041X. DOI: 10.1007/s40072-025-00384-x. arXiv: 2401.05935 [math.PR]. URL: http://arxiv.org/abs/2401.05935 (visited on 06/15/2026).